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EDUCATION

Sankalchand Patel University
Bachelor of Technology in Computer Engineering (CGPA: 7.80 / 10.0)

June 2023 – June 2027
Visnagar, Gujarat, India

- Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Deep Learning, Database Management Systems (DBMS), Operating Systems, Software Engineering, Object-Oriented Programming, Computer Networks.

TECHNICAL SKILLS

Languages & Backend:
AI & Agent Systems:

Python (Core), C++, SQL, JavaScript, TypeScript, FastAPI, Flask, RESTful APIs, Pydantic
LangGraph, LangChain, Autonomous Agents, Tool Calling, LLaMA 3.3 70B, OpenAI API, Groq, PyTorch

Retrieval & Databases:
Developer Tools & DevOps:

RAG Pipelines, Self-RAG, FAISS, Qdrant, Pinecone, PostgreSQL, Redis, SQLite, MongoDB
Docker, Git, GitHub, Vercel, Render, Postman, UptimeRobot, Linux/Bash

FEATURED PROJECTS

AgentPay | *Python, FastAPI, LangGraph, PostgreSQL, Redis, Next.js, Razorpay API*

Feb 2026

- Engineered autonomous buyer-agent infrastructure where the LLM proposes purchases, but an isolated **deterministic policy engine gates every payment action** before invoking Razorpay’s API, eliminating unconstrained agent execution.
- Enforced an append-only, **database-trigger-backed audit trail** tracking every state transition, tenant merchant isolation, and hard consumer spend-authorization limits.
- Implemented a sandbox-to-production compliance pipeline with webhook-driven agent integration, **HMAC signature verification**, and automated retry backoff handling.

CampusCircle | *Next.js, FastAPI, PostgreSQL, LLaMA 3.3 70B, TypeScript*

Nov 2025 – Jan 2026

- Architected and deployed the full backend and AI infrastructure for a verified student platform, scaling solo to **200+ active campus users** with sub-150ms average API response times.
- Shipped 3 production AI features powered by **LLaMA 3.3 70B**: an automated support agent (“Reva”), an adaptive quiz engine dynamically tuning question difficulty, and a YouTube-to-structured-study notes summarizer.
- Awarded **Top 18 of 500+ participants** nationwide in The Maverick Ect AI Challenge (Dewang Mehta Foundation Trust).

CliniqueAI | *Python, Flask, PostgreSQL, Scikit-learn, XGBoost, LangChain, Pydantic*

Jan 2026 – Feb 2026

- Trained and benchmarked Logistic Regression, Random Forest, and **XGBoost classifiers** for clinical diabetes risk prediction, optimizing the decision boundary for medical explainability and high recall.
- Integrated a **RAG patient copilot** grounded on localized EHR records, transactional CRUD endpoints, Pydantic data schemas, and physician notification alerts.
- Eliminated cloud cold-start latency on Render free tier by configuring an automated heartbeat via UptimeRobot.

RAG Architectures Lab | *Python, LangChain, FAISS, Qdrant, Pinecone, FastAPI, Streamlit*

2025 – 2026

- Implemented a comparative playground for RAG paradigms, benchmarking naive dense retrieval vs. conversational memory vs. self-corrective agentic loops with automated query reformulation.
- Evaluated vector search indexing latency and precision tradeoffs across Qdrant, Pinecone, and in-memory FAISS.

HONORS & CERTIFICATIONS

National AI Hackathon (Unstop) — Top 20 Finalist (out of 1,600+)
Selected among top 20 software teams nationwide for developing CliniqueAI diagnostic copilot

Feb 2026
National Level

The Maverick Ect AI Challenge — Top 18 Finalist (out of 500+)
Recognized by Dewang Mehta Foundation Trust for building CampusCircle AI community platform

2026
National Level

HackerRank Certified Software Engineer
Demonstrated competency in Problem Solving, SQL Database Queries, and RESTful API Architecture

2025
Verified Certificate